

Where the runtime goes, compression OFF vs ON: every second of the run is attributed
QuEST GPU+MPI, 30 qubits / 4 ranks, 4×RTX 2080 Ti (landonia01); slowest of the 4 ranks shown — it sets the wall time
Compression mainly removes communication time; fast ON bars expose the roughly constant lifecycle start-up.
Top: seconds. Bottom: the same runs as shares of wall time. 'Execution overhead' is gate-execution time outside computation and communication; 'unclassified residual' is wall time outside all categories — 0.0 s in all panels, so the attribution is complete.
Lifecycle is approximately constant environment start-up (4–6 s), so its share grows when compression shortens the run.

